

45.6 Comparing distributional models

Three fits to the same data now sit side by side: a homoscedastic linear model, a family of quantile regressions, and a location-scale GAMLSS. They estimate different things, so none can be judged by how well it hits its own target. What they share is that each delivers, at every covariate value, a predictive distribution for the response. That is the object to score.

45.6.1 Scoring rules

Definition 45.6.1 — Scoring rule, propriety

A **scoring rule** is a function $S(F, y)$ assigning a penalty when the predictive distribution F is issued and the value y is observed; smaller is better. Write $S(F, G) = \mathbb{E}_{Y \sim G} S(F, Y)$ for its expectation when Y really follows G . The rule is **proper** relative to a class of distributions if

$$S(G, G) \leq S(F, G) \quad \text{for all } F, G \text{ in the class,} \quad (45.6.1)$$

and **strictly proper** if equality forces $F = G$. A **consistent scoring function** for a functional T is a function $s(x, y)$ with $\mathbb{E}_G s\{T(G), Y\} \leq \mathbb{E}_G s(x, Y)$ for every x ; a functional admitting one is called **elicitable**.

Propriety is the minimal honesty requirement: under a rule that is not proper a forecaster does better by misreporting, and a comparison of methods rewards whoever misreports most skilfully. Three consistent scoring functions have already appeared: squared error for the mean, the check loss for the τ -quantile by Theorem 45.2.1(a), and Eq. 45.4.1 for the τ -expectile by Proposition 45.4.1(a). Each scores a single number; the next result scores a whole distribution and ties it back to the check loss.

Proposition 45.6.1 — The continuous ranked probability score and the probability integral transform

For a distribution function F with finite mean and a real y , define

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} \{F(s) - \mathbf{1}\{y \leq s\}\}^2 ds. \quad (45.6.2)$$

(a) $\text{CRPS}(F, y)$ is finite, and for any G with finite mean

$$\mathbb{E}_{Y \sim G} \text{CRPS}(F, Y) = \int \{F(s) - G(s)\}^2 ds + \int G(s) \{1 - G(s)\} ds. \quad (45.6.3)$$

Hence the CRPS is a strictly proper scoring rule on the distributions with finite mean.

(b) If F is continuous and strictly increasing with finite mean, then

$$\text{CRPS}(F, y) = 2 \int_0^1 \rho_\tau \{y - F^{-1}(\tau)\} d\tau : \quad (45.6.4)$$

the CRPS is the check loss averaged over all levels.

- (c) If $Y \sim F$ with F continuous, then $U = F(Y)$ is uniform on $(0, 1)$. If the predictive distributions $F_1, \dots, F_n$ are the true conditional distributions of independent $Y_1, \dots, Y_n$ and each is continuous, then the **probability integral transforms** $U_i = F_i(Y_i)$ are independent and uniform.

Proof.

- (a) The integrand of Eq. 45.6.2 is bounded by one and vanishes outside the region where $F(s)$ and $\mathbf{1}\{y \leq s\}$ disagree, which lies in the set where $\min\{F(s), 1 - F(s)\} > 0$ together with a bounded interval; a finite mean makes $\int \min\{F, 1 - F\} ds < \infty$, so the score is finite. For fixed s , $\mathbf{1}\{Y \leq s\}$ is Bernoulli with mean $G(s)$, so

$$\mathbb{E}_G \{F(s) - \mathbf{1}\{Y \leq s\}\}^2 = \{F(s) - G(s)\}^2 + G(s)\{1 - G(s)\}.$$

Integrating over s (Tonelli, the integrand being nonnegative) gives Eq. 45.6.3. The second term does not involve F , so the expectation is minimized exactly when $\int (F - G)^2 = 0$, that is when $F = G$ almost everywhere, and two right-continuous distribution functions agreeing almost everywhere are equal.

- (b) Write $q_\tau = F^{-1}(\tau)$ and $u = F(y)$, and split both sides at y . For the left side, $F(s) - \mathbf{1}\{y \leq s\}$ equals $F(s)$ for $s < y$ and $F(s) - 1$ for $s \geq y$, so $\text{CRPS}(F, y) = \int_{-\infty}^y F^2 + \int_y^\infty (1 - F)^2$. Substituting $s = q_\tau$ and integrating by parts,

$$\begin{aligned} \int_{-\infty}^y F(s)^2 ds &= \int_0^u \tau^2 dq_\tau = u^2 y - 2 \int_0^u \tau q_\tau d\tau, \\ \int_y^\infty \{1 - F\}^2 ds &= 2 \int_u^1 (1 - \tau) q_\tau d\tau - (1 - u)^2 y, \end{aligned}$$

the boundary terms at 0 and 1 vanishing because the mean is finite. For the right side, $q_\tau \leq y$ exactly when $\tau \leq u$, so $\rho_\tau(y - q_\tau) = \tau(y - q_\tau)$ for $\tau < u$ and $(1 - \tau)(q_\tau - y)$ for $\tau > u$. Hence

$$\begin{aligned} 2 \int_0^1 \rho_\tau(y - q_\tau) d\tau &= 2y \int_0^u \tau d\tau - 2 \int_0^u \tau q_\tau d\tau \\ &\quad + 2 \int_u^1 (1 - \tau) q_\tau d\tau - 2y \int_u^1 (1 - \tau) d\tau, \end{aligned}$$

and $2 \int_0^u \tau d\tau = u^2$, $2 \int_u^1 (1 - \tau) d\tau = (1 - u)^2$. The two expressions agree term by term.

- (c) For continuous F and $u \in (0, 1)$, the set $\{y : F(y) \leq u\}$ is the interval $(-\infty, q_u]$ up to endpoints of flat stretches, which carry no probability, so $P\{F(Y) \leq u\} = F(q_u) = u$ by continuity. Independence of the U_i follows from independence of the Y_i , each F_i being a fixed function once the covariates are given.

■

Part (b) is the bridge between the two halves of this chapter. A quantile regression fitted at a dense grid of levels gives a predictive quantile function, evaluated by Eq. 45.6.4 with exactly the loss it was fitted with; a GAMLSS gives a predictive distribution function, evaluated by Eq. 45.6.2. The same number, in the units of the response, is available for both.

Remark (Other proper rules). The **logarithmic score** $S(F, y) = -\log f(y)$ is strictly proper and is what likelihood fitting optimizes, but it needs a density and is unbounded: one observation where the predictive density is nearly zero dominates everything, as Example (Watching it happen) showed. The CRPS needs no density, is finite for every observation, and grows only linearly in how far the observation falls from the forecast where the log score diverges — the safer default across methods. For a single level, the **pinball loss** $\rho_\tau(y - \hat{q}_\tau)$ reports how well one quantile is predicted.

45.6.2 Calibration and sharpness

Part (c) gives the standard graphical check. Compute $u_i = \hat{F}_i(y_i)$ for held-out observations and plot their histogram: under a correct model it is flat, and departures have a vocabulary. A **hump** in the middle means predictive distributions that are too wide; a **U shape** too narrow, the common and dangerous failure; a **slope** a biased location; **spikes at 0 and 1** alone, wrong tails with a sound body.

For a discrete response $F_i(Y_i)$ is not uniform, and the histogram must use a randomized version, drawing U_i uniformly on $[F_i(y_i^-), F_i(y_i)]$: the randomized quantile residuals of Dunn and Smyth (1996), exactly normal under a correct model after applying Φ^{-1} .

Warning (In-sample calibration proves nothing). The transforms must be computed out of sample. A fitted model is calibrated on its own training data almost by construction: Proposition 35.4.2 showed that in a logistic regression with an intercept the recalibration slope is exactly one and the intercept exactly zero, whatever the model's merits. A location-scale fit that has just matched $\hat{\sigma}$ to its own residuals gives a flat in-sample histogram whether or not its scale curve is right.

Calibration alone is not enough either: the unconditional distribution of the response is perfectly calibrated and useless. The principle of Gneiting, Balabdaoui and Raftery (2007) is to maximize **sharpness subject to calibration** — among predictive distributions that pass the check, prefer the most concentrated. A proper scoring rule does both at once, which is why the scores come first.

45.6.3 Three models, one comparison

Example. (Mean model, quantile model and GAMLSS on Engel's budgets)

The 235 households are split into ten folds. On each fold, three predictive distributions are built from the other nine:

- (i) the homoscedastic linear model of Part II, with normal errors and one $\hat{\sigma}$;
- (ii) quantile regressions at 199 levels $\tau = 1/200, \dots, 199/200$, sorted at each covariate value by Proposition 45.3.3(b) so that the fitted quantiles are ordered;

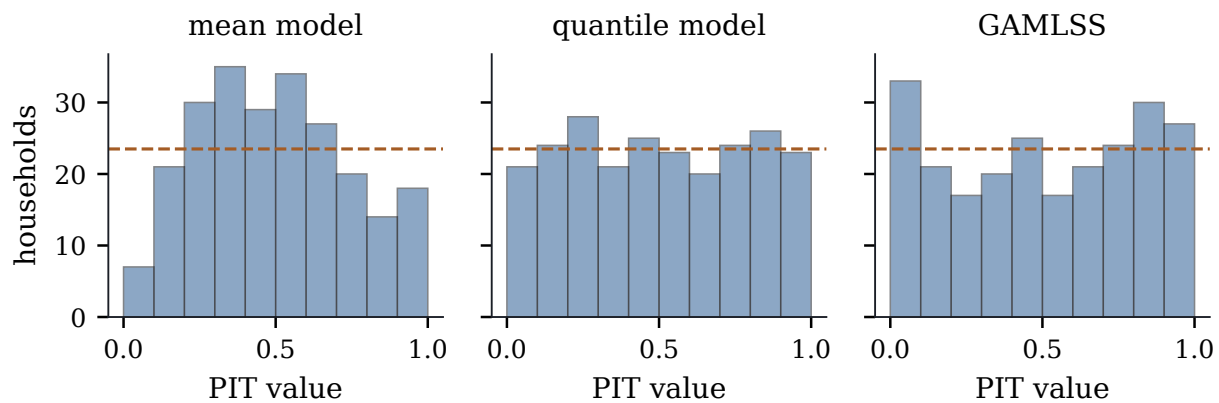


Figure 45.6.1: Held-out probability integral transforms from ten-fold cross-validation; a correct model gives a flat histogram (dashed). The mean model is humped, the quantile model close to flat, the GAMLSS raised at both ends.

(iii) the location-scale GAMLSS of Example (Mean and scale together).

All three are scored the same way, by the discretized form of [Eq. 45.6.4](#) over that grid of levels, so no model is helped by a formula the others do not get.

model	CRPS	pinball 0.1	pinball 0.5	pinball 0.9	below $\hat{q}_{0.1}$	below $\hat{q}_{0.9}$
mean model	59.34	20.41	38.98	21.76	0.030	0.923
quantile model	53.00	17.69	38.05	14.93	0.089	0.902
GAMLSS	53.12	16.26	37.53	16.49	0.140	0.885

Scores are in francs, and each model gets its own best predictive distribution: for the mean model the exact t predictive of [Section 12.3](#). Both distributional models beat it by about 6 francs of CRPS per household: the differences are -6.35 with a paired standard error of 1.75, and -6.23 with standard error 2.12. Against each other they differ by -0.12 with a paired standard error of 1.99, which is no difference at all. The mean model fails where Proposition 45.1.2 predicted, putting only 0.030 of the households below its own tenth percentile.

The Kolmogorov–Smirnov distance of the held-out transforms from uniformity is 0.098 for the mean model, 0.024 for the quantile model and 0.055 for the GAMLSS ([Figure 45.6.1](#)). The mean model’s hump is the signature of predictive distributions that are too wide; the GAMLSS histogram is raised at both ends, the signature of a normal shape where the data want heavier tails — an actionable criticism, whose answer is a three-parameter family.

```
import numpy as np
from scipy import stats

LEVELS = np.arange(1, 200) / 200.0          # the tau grid the CRPS integral uses

def check_loss(u, tau):
    """The check function of Section 45.2."""
```

```

    return u * (tau - (u < 0))

def crps_from_quantiles(q, y):
    """CRPS = 2 * integral of the check loss over tau, on the grid LEVELS.

    q has one row per observation and one column per level of LEVELS.
    """
    return 2 * np.mean(check_loss(y[:, None] - q, LEVELS[None, :]), axis=1)

def pit_from_quantiles(q, y):
    """The probability integral transform read off a set of fitted quantiles."""
    return np.mean(q <= y[:, None], axis=1)

# one household: a normal predictive with mean 600 and scale 120 francs, observed at
# 700
q_one = 600.0 + 120.0 * stats.norm.ppf(LEVELS[None, :])
y_one = np.array([700.0])
print(f"CRPS {crps_from_quantiles(q_one, y_one)[0]:.2f} francs,"
      f" PIT {pit_from_quantiles(q_one, y_one)[0]:.3f}")

```

[Run this code online](#)

45.6.4 Reading a score difference honestly

Three cautions close the comparison, the same three that have followed every model comparison in this book. A difference needs a standard error: 59.34 against 53.00 is three and a half paired standard errors, 53.00 against 53.12 less than a tenth of one, and the formal test, corrected for dependence in time series, is Diebold and Mariano (1995). Selecting on the score inflates it: the GAMLSS smoothing parameters were chosen in Example (Mean and scale together) on these same ten folds, so its score is optimistic in the way Proposition 29.5.1 describes. And a score is not a purpose: the CRPS weights all levels alike, whereas an insurer who cares about the upper tail should compare pinball losses at $\tau = 0.95$ and a nutritionist concerned with the poorest households at $\tau = 0.05$. Decide what the model is for, then choose a scoring function consistent for it.

45.6.5 Exercises

45.6.5.1 A. Check your understanding

Exercise A1.

A forecaster issues $N(0, 4)$ every day while the truth is $N(0, 1)$. Sketch the probability integral transform histogram, and say which of the four shapes listed above appears.

Solution. $U = \Phi(Y/2)$ with $Y \sim N(0, 1)$ concentrates near $1/2$, since $Y/2$ rarely leaves $(-1, 1)$ and Φ maps that to $(0.16, 0.84)$: a hump in the middle, the signature of predictive distributions that are too wide.

45.6.5.2 B. Practice

Exercise B1.

Show that for $F = N(\mu, \sigma^2)$ and $z = (y - \mu)/\sigma$,

$$\text{CRPS}(F, y) = \sigma \left\{ z(2\Phi(z) - 1) + 2\varphi(z) - \frac{1}{\sqrt{\pi}} \right\},$$

using the identity $\text{CRPS}(F, y) = E|X - y| - \frac{1}{2} E|X - X'|$ for independent $X, X' \sim F$.

Solution. For $Z \sim N(0, 1)$ and a constant a , integrating by parts gives $E|Z - a| = 2\varphi(a) + a\{2\Phi(a) - 1\}$. Writing $X = \mu + \sigma Z$, $E|X - y| = \sigma E|Z - z| = \sigma\{2\varphi(z) + z(2\Phi(z) - 1)\}$. Next $X - X' \sim N(0, 2\sigma^2)$, and $E|N(0, v^2)| = v\sqrt{2/\pi}$, so $E|X - X'| = \sigma\sqrt{2} \cdot \sqrt{2/\pi} = 2\sigma/\sqrt{\pi}$. Substituting both into the identity gives the stated formula. By [Proposition 45.6.1\(a\)](#) the expected score is minimized in (μ, σ) at the true values.

Exercise B2.

Prove the identity used in [Exercise \(B1\)](#): $\text{CRPS}(F, y) = E|X - y| - \frac{1}{2} E|X - X'|$ for independent $X, X' \sim F$ with finite mean.

Solution. The key is $|a - b| = \int \{\mathbf{1}\{a \leq s\} - \mathbf{1}\{b \leq s\}\}^2 ds$, since the integrand is one exactly on the interval between a and b . Taking $a = X$, $b = y$ and expectations, and using $E\{A - c\}^2 = \text{Var}(A) + (E A - c)^2$ with $A = \mathbf{1}\{X \leq s\}$,

$$E|X - y| = \int \left[\{F(s) - \mathbf{1}\{y \leq s\}\}^2 + F(s)\{1 - F(s)\} \right] ds = \text{CRPS}(F, y) + \int F(1 - F) ds.$$

Applying the same identity to the independent pair, $E\{A - A'\}^2 = 2 \text{Var}(A)$ gives $E|X - X'| = 2 \int F(1 - F) ds$. Subtracting half of the second display from the first gives the identity; finiteness of the mean makes $\int F(1 - F) ds$ finite.

45.6.5.3 C. Going deeper**Exercise C1.**

Interpret the two terms of [Eq. 45.6.3](#). Show that the second depends only on the truth G , so it is an irreducible floor, and that the first is an integrated squared error between the forecast and the truth. Relate this to the bias-variance decomposition of prediction error in [Section 29.1](#).